low latency, high availability, or localized decision-making often benefit from a distributed authorization model. In this model, each application or service includes its own authorization client—a bundled component that manages authorization requests locally by integrating the Policy Enforcement Point (PEP), Policy Decision Point (PDP), and a local entity cache. This setup is illustrated in figure 4.7.

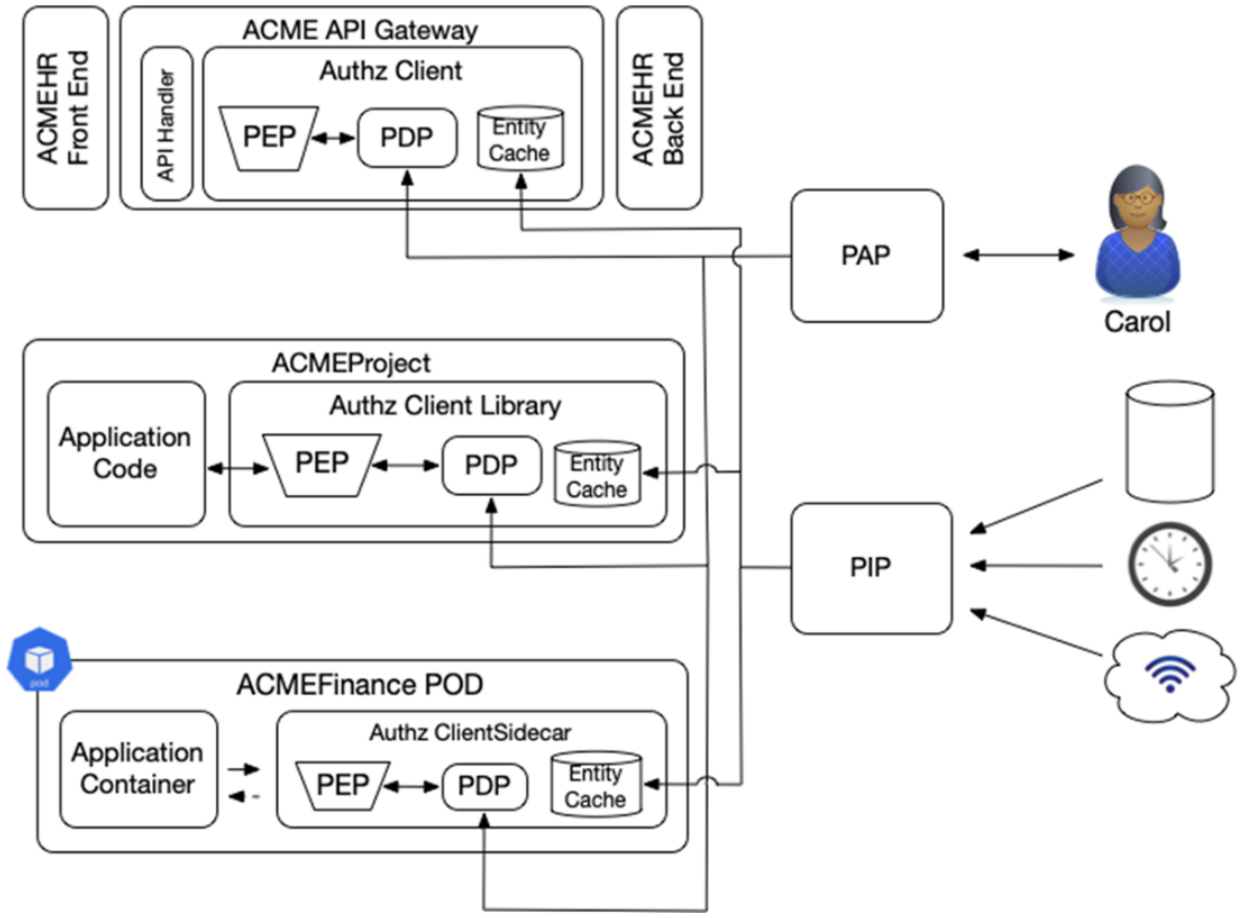


Figure 4.7 Distributed authorization architecture at ACME Corp. Each service includes an authorization client—a bundled component that contains the PEP, PDP, and an entity cache. The client may be deployed as a sidecar (ACMEFinance), as a library within application code (ACMEProject), or integrated into an API gateway (ACMEHR). Policies are authored and managed by Carol via the Policy Administration Point (PAP) and then distributed to each client. Contextual data is retrieved from shared Policy Information Points (PIPs), such as identity stores, time servers, and network telemetry, and is cached inside the client.

These authorization clients can be deployed in the same ways that the PEP was deployed, depending on the architecture of the service. At ACME Corp, they appear as: